

COMP.2560: System Programming

LAB #9: Named Pipes [FIFO]

LAB 9: Overview

In this lab we will explore Named Pipes, a.k.a. FIFO.

There are three parts in this lab, described below.

LAB 9 – Part I (Question #11)

Study the `sendmessage.c` and `rcvmessage.c` source code posted.

In `rcvmessage.c`, the FIFO is opened using the “O_RDWR” mode. Change the FIFO open mode from O_RDWR to O_RDONLY to see what happens to this pair of programs. If nothing different happens, why would one prefer to use O_RDWR instead of using O_RDONLY?

No coding is required but you are expected to write a short paragraph to answer the questions above. Submission is required for this question at the end of your lab session.

LAB 9 – Part II (Question #2)

Google online, read the man page, or read relevant pages in our textbook (use the index of the textbook) regarding signal **SIGPIPE**.

Design a small program in which **SIGPIPE** is generated when using FIFO and caught by your program (This is similar to a question in an assignment.).

Please code the program described above. Make sure you add comments, explaining your work. Submission is required for this question at the end of your lab session.

LAB 9 – Part III (Question #3)

We studied the source code in file `game.c` showing a referee coordinating two players to play a simple game of each player in turn rolling dice and accumulating points to see who first gets 50 points. This program is for two players only.

Modify the `game.c` program so that there are **3 players** in the game and the referee will now need to coordinate the **3 players** to play the game in turn.

Please code the program described above. Make sure you add comments, explaining your work. Submission is required for this question at the end of your lab session.

LAB 9 – Submission

Submit the items elements described above.

To formally submit your work for this Lab:

- Submit the code (including extended commenting) and the script files (learned in Lab 2) you created for Parts II, and III.
- Submit the paragraph you prepared for Part I.

When submitting, please indicate in the submission textbox the Lab section you belong to.

Evaluation:

1. It is our intention to evaluate your work being done in-person, while you are in the lab.
2. It is strongly suggested that you attend in-person to familiarize yourselves with the Lab infrastructure, meet with the GATAs and solve problems.
3. Each session is only 80 minutes long, so it is strongly suggested that you arrive early, are focused and do not disrupt the process.
4. It is desired that you complete your work while in the lab and get graded after live review from the GATAs.
5. Should you need more time, you can take your work home and present it to the GATAs on the very next lab session (the week following).
6. Even if you decide to work from home and submit your work remotely, you still need to have your work peer reviewed, in-person.